

Module Review: Osmoregulated Periplasmic Glucan Biosynthesis (UniPathway UPA00637) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module/bucket:** unipathway:UPA00637 — "Glycan metabolism; osmoregulated periplasmic glucan (OPG) biosynthesis" **Candidate genes:** opgH (PP_5025 / Q88D04), opgG (PP_5026 / Q88D03)

1. Executive summary

- The core **OPG backbone-synthesis module (UPA00637)** is **satisfiable and should be marked COVERED** in KT2440. Both defining genes are present, adjacent, and correctly annotated: **opgH (PP_5025, inner-membrane GT2 glucosyltransferase)** and **opgG (PP_5026, periplasmic OpgG/MdoG-family maturation protein)**.
- Evidence for KT2440 itself is **homology-based (UniProt protein existence level 3, "inferred from homology")**, but transfer is **strong** because the direct orthologs in the close congener *P. aeruginosa* (opgGH, PA5077/PA5078) are experimentally shown to make linear β -1,2 glucan (Lequette et al. 2007,

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), and opgG/opgH are the most conserved, functionally validated OPG genes across proteobacteria.

- **No accessory OPG genes** (opgB/mdoB phosphoglycerophosphotransferase, opgC/mdoC succinyltransferase, opgD glucan hydrolase, or the non-homologous ndvB) are annotated in the KT2440 proteome. These are **decoration/tailoring steps outside the UPA00637 backbone bucket** and should be treated as `not_expected_in_target_taxon` **for this module** (their true presence is a separate question — see §5).

- Curation risks are limited to **broad EC/GO breadth on OpgH** (bare EC 2.4.1.-, generic GT2 Pfam) and an **unannotated moonlighting FtsZ/cell-size role** documented for *E. coli* OpgH.

2. Target-organism pathway definition

Process included (UPA00637): Synthesis of the **glucose backbone of osmoregulated periplasmic glucans (OPGs)** — cytoplasmic-facing polymerization of UDP-glucose-derived glucosyl units by the inner-membrane glucosyltransferase **OpgH**, coupled to **OpgG**-dependent periplasmic maturation/assembly of the β -1,2-linked (with β -1,6 branch) glucan. OPGs accumulate in the periplasm in response to **low medium osmolarity** and are envelope homeostasis molecules.

Alternate names / database definitions (keep consistent): - OPGs = historically "**membrane-derived oligosaccharides (MDOs)**"; *E. coli* genes **mdoG/mdoH** = **opgG/opgH**. - UniPathway UPA00637 $\equiv$ backbone synthesis only.

Neighboring pathways/maps to keep SEPARATE: - **OPG substitution/decoration** (glycerophosphotransferase OpgB, succinyltransferase OpgC) — modifies but does not build the backbone. - **Periplasmic glucan tailoring/hydrolysis** (OpgD) — controls chain length. - **Cyclic β -1,2 glucan synthesis via NdvB/ChvB** (Rhizobiales/*Agrobacterium*) — a **non-homologous, alternative OPG route**; not the opgGH route. - **Glycogen / α -1,4-glucan metabolism** (KT2440 glgB/PP_4058, glgE/PP_4060, glgP/PP_5041, glgA/glgC) — cytoplasmic storage glucan, unrelated to periplasmic OPGs. Do **not** merge these into UPA00637 despite the shared "glucan" keyword. - Broad "starch and sucrose metabolism" / KEGG overview maps — keep out of the module.

3. Expected step model

Step	Enzyme/ player	Role	Expected in KT2440?
1. Backbone polymerization	OpgH (inner-membrane GT2)	Glucosyltransferase; polymerizes β -1,2 glucan backbone from UDP-glucose	Yes — covered (PP_5025)
2. Periplasmic maturation/assembly	OpgG (periplasmic)	Backbone maturation / branching-associated periplasmic factor; carbohydrate binding	Yes — covered (PP_5026)
(accessory) Glycerophosphotransferase	OpgB/MdoB	sn-1-P-glycerol substitution	Not in module scope; not annotated
(accessory) Succinyltransferase	OpgC/MdoC	Succinyl substitution (seen in <i>P. aeruginosa</i> OPGs)	Not in module scope; not annotated (open question)
(accessory) Glucan hydrolase	OpgD	Chain-length control	Not in module scope; not annotated

Both required backbone steps are encoded. Accessory steps are **not part of UPA00637** and their absence does not reduce module satisfiability.

4. Candidate genes and evidence

opgH — PP_5025 / Q88D04 (high confidence; COVERS step 1)

- **Annotation:** Glucans biosynthesis glucosyltransferase H; 857 aa; **Cell inner membrane** (multi-TM); UniPathway UPA00637.
- **Family evidence:** InterPro **IPR023725 (Glucans_biosynth_gluTrFase_H, OpgH-specific)**, IPR050321 (Glycosyltr_2/OpgH_subfam), generic GT2 IPR001173/Pfam PF00535; **COG2943**.

The OpgH-specific InterPro + COG + pathway context make the subfamily call reliable despite the generic GT2 core.

- **Function transfer:** Strong. Orthologous *P. aeruginosa* opgH (PA5077) participates in experimentally verified linear β -1,2 glucan synthesis

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mdoH/opgH mutants abolish OPGs in *E. coli*, *Erwinia*, *Salmonella*, *Shigella*, *Rhodobacter* (PMIDs 11325942, 19118363, 20062978, 12027885).

- **Curation caveats:** (i) EC is only 2.4.1.- (no leaf EC) and the generic GT2 Pfam/GO are **over-propagation-prone** — rely on the OpgH-subfamily anchor. (ii) *E. coli* OpgH **moonlights** as a UDP-glucose-dependent, nutrient-sensing FtsZ antagonist controlling cell size (Hill et al. 2013,

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this second function is **not annotated** in KT2440.

opgG — PP_5026 / Q88D03 (high confidence; COVERS step 2)

- **Annotation:** Glucans biosynthesis protein G; 559 aa; **Periplasm** with cleaved signal peptide; UniPathway UPA00637.
- **Family evidence:** **OpgD/OpgG family**; Pfam **PF04349 (MdoG)**, InterPro IPR023704 (MdoG_OpgG), IPR007444 (MdoG_C), galactose-mutarotase-like + Ig/E-set folds; **COG3131**.
- **Function transfer:** Strong (same experimental basis as opgH; opgG mutants abolish OPGs).
- **Curation caveats:** OpgG and OpgD are **paralogs of the same family** (PF04349); automated pipelines can mis-split OpgG vs OpgD. **Verified: KT2440 has exactly one PF04349 member (PP_5026), i.e. no separate OpgD**, so the periplasmic role is assigned to **OpgG with no paralog ambiguity** in this genome. GO carbohydrate-binding term is appropriate; avoid asserting a specific hydrolase activity absent OpgD-type evidence.

GT2-breadth check: OpgH (PP_5025) is **1 of 7 GT2/PF00535 proteins** in KT2440 (others: PP_3256, PP_1792, PP_1793, PP_3137, PP_3135, PP_0034), but **only PP_5025 carries the OpgH-specific InterPro IPR023725** — concretely confirming that the shared GT2 Pfam/GO is over-propagation-prone and that the specific OpgH call depends on the subfamily/UniPathway anchor.

Operon context (verified): PP_5025/PP_5026 are consecutive, matching the conserved **opgHG (opgGH) operon** organization (as in *P. aeruginosa* PA5077/PA5078). The locus is a **standalone two-gene unit**: upstream neighbors are an amino-acid ABC-transport cluster (PP_5023 GlnP

permease, PP_5024 solute-binding protein 3 family), downstream are unrelated genes (PP_5027 D-aminoacyl-tRNA deacylase, PP_5028 proline iminopeptidase). **No adjacent opgB/opgC/opgD acyl-/glycerophosphotransferase** — no genomic evidence for a co-localized OPG-decoration sub-module.

5. Gaps, ambiguities, and likely over-annotations

- **Backbone module: no gaps.** Both required steps covered by high-confidence genes.
 - **Accessory tailoring genes absent from annotation:** No opgB, opgC, opgD, or ndvB found in a proteome-wide search of taxon 160488. *P. aeruginosa* OPGs carry succinyl substituents, so an **opgC-like succinyltransferase could plausibly exist** but be unannotated/divergent — a genuine open question, though **outside UPA00637**. Mark accessory steps `not_expected_in_target_taxon` for this module while noting the opgC uncertainty for a possible substitution sub-module.
 - **Over-annotation risk on OpgH:** bare `EC 2.4.1. -` and generic GT2 GO terms; keep the specific OpgH-subfamily/UniPathway anchoring.
 - **Do not conflate glycogen genes** (glgB/E/P/A/C) with OPG biosynthesis; shared "glucan" wording is the main false-merge risk.
 - **Evidence class:** All KT2440 assignments are homology-inferred (PE3); there is **no direct KT2440 experimental OPG characterization** in the retrieved literature.
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6. Module and GO-curation recommendations

Module step	Recommendation
OpgH glucosyltransferase step	covered (PP_5025)
OpgG periplasmic maturation step	covered (PP_5026)
Overall UPA00637 backbone module	SATISFIED / covered in KT2440
OPG succinylation (opgC)	<code>not_expected_in_target_taxon</code> for UPA00637; flag as <code>candidate_uncertain</code> if a substitution sub-module is defined
OPG glycerophosphate (opgB), chain-length (opgD)	<code>not_expected_in_target_taxon</code> for this backbone module

- **Module boundaries are appropriate** for KT2440 as a backbone-only (OpgH + OpgG) module; no revision needed. Keep decoration/hydrolysis and cyclic-glucan (ndvB) routes as separate modules.
- **GO:** OpgH — retain glucosyltransferase activity + OPG biosynthetic process + inner membrane; consider (only if experimentally supported in *Pseudomonas*) an added division/FtsZ-regulation term. OpgG — carbohydrate binding + OPG biosynthetic process + periplasm; avoid a specific hydrolase term. No new GO term requests appear necessary for the core module.

7. Genes to promote to full `fetch-gene` review

1. **opgH / PP_5025 (Q88D04)** — promote. Reasons: broad `EC 2.4.1.-`, generic GT2 GO breadth, and a documented moonlighting FtsZ/cell-size function worth curating or explicitly excluding for KT2440.
2. **opgG / PP_5026 (Q88D03)** — promote (lower priority). Confirm OpgG-vs-OpgD family assignment is unambiguous (single PF04349 member) and that carbohydrate-binding/periplasmic annotation is correctly scoped.

8. Key references

- Bohin JP. 2000. Osmoregulated periplasmic glucans in Proteobacteria.

P 10779706

(review; OPG families, osmoregulation, pleiotropic mutant phenotypes).

- Lequette Y, Rollet E, Delangle A, Greenberg EP, Bohin JP. 2007. Linear osmoregulated periplasmic glucans are encoded by the *opgGH* locus of *Pseudomonas aeruginosa*.

P 17906125

(direct *Pseudomonas* evidence: β -1,2 backbone, β -1,6 branches, succinyl substitution, osmotic repression, biofilm role).

- Hill NS, Buske PJ, Shi Y, Levin PA. 2013. A moonlighting enzyme links *E. coli* cell size with central metabolism.

P 23935518

(OpgH as UDP-glucose-dependent FtsZ antagonist / cell-size regulator).

- Page F et al. 2001. OPG synthesis required for *Erwinia chrysanthemi* pathogenicity.

P 11325942

(*opgGH* $\equiv$ *mdoGH*; mutants abolish OPGs).

- Bhagwat AA et al. 2009. OPGs of *Salmonella* Typhimurium required for virulence.

P 19118363

(bicistronic *opgGH* operon).

- Liu D et al. 2010. OPG gene family of *Shigella flexneri*.

P 20062978

(*opgGH* + accessory *opgB/C/D* operon structure).

- Cogez V et al. 2002. *opgGIH/opgC* operon in *Rhodobacter sphaeroides*.

P 12027885

(backbone vs succinylation genetics).

- UniProt Q88D04 (*opgH*/PP_5025) and Q88D03 (*opgG*/PP_5026); UniPathway UPA00637; InterPro IPR023725, IPR023704, Pfam PF00535/PF04349.

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